[Metric]
A-A-2331A
November 20, 1995
Superseding
A-A-2331
October 1, 1985

COMMERCIAL ITEM DESCRIPTION
INK, DRAWING
(WATERPROOF, COLORED)

The General Services Administration has authorized the use of this commercial item description, by all federal agencies.

1. Scope: This Commercial Item Description covers colored, waterproof drawing ink for general drafting use (drawing and tracing).

2. Classification:

Type	I.	Transparent
	II.	Opaque

3. Salient Characteristics:

3.1 Type I Transparent Ink, shall be a transparent solution of a dye in a liquid medium.

3.2 Type II Opaque Ink, shall be a combination of transparent dye in solution and opaque pigment in suspension in a liquid medium.

3.3 The vehicle of inks shall be an aqueous solution of the necessary ingredients to prevent the precipitation of the dye or the separation of the pigment and to leave a waterproof film on drawing paper.

3.4 Colors of ink furnished shall be as listed in Table I, as specified.

3.5 The coloring matter shall be dyes or pigments of suitable color and brilliance, with permanence to light. The coloring matter shall be dyes or pigments of suitable color and brilliance, with permanence to light as specified in paragraph 5.3.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to General Services Administration, Federal Supply Service, Engineering Branch, 26 Federal Plaza, New York, NY 10278.

FSC 7510

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3.6 Stability. The ink shall not show any coagulation, granulation, or precipitation under working or storage conditions. Such evidence of instability shall not be present when the ink is received or after one year of storage under normal conditions.

3.7 Colors (formulas). Inks of all colors (table I) shall be equal to inks prepared according to the following formula:

To prepare 100 mm of ink, dissolve a quantity of dye as indicated in table I for the respected colors, in 50 mm of warm water and add 50 mm of shellac solution (100 grams of wax-free, bleached, white shellac per liter of water). Cool to room temperature and filter.

TABLE I
Drawing Ink Color Control Formulas

COLORS	COLOR INDEX	GRAINS OF DYE
	TRANSPARENT INKS	
Blue	C.I. 44095	
	C.I. Acid Blue 97	0.5000
Brown	C.I. 30045	
	C.I. Direct Brown 1	0.8000
Carmine	C.I. 18070	
	C.I. Acid Red 40	0.6400
	C.I. 27290	
	C.I. Acid Red 73	0.1600
Green	C.I. 42100	
	C.I. Acid Green 9	1.2000
Indigo	C.I. 42735	
	C.I. Acid Blue 104	0.7350
	C.I. 20470	
	C.I. Acid Black 1	0.4350
	C.I. 45380	
	C.I. Acid Red 87	0.0300
Orange	C.I. 16020	
	C.I. Acid Orange 17	0.6000
Red	C.I. 45425	
	C.I. Acid Red 95	0.5000

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TABLE I. CONT.

Red Brick	C.I. 27290	
	C.I. Acid Red 73	0.3500
	C.I. 13065	
	C.I. Acid Yellow 36	0.1200
	C.I. 42460	
	C.I. Acid violet 49	0.0060
Red Violet	C.I. 45440	
	C.I. Acid Red 94	0.6487
	C.I. 42640	
	C.I. Acid Violet 49	0.1012
Russet	C.I. 21010	
	C.I. Basic Brown 4	0.5000
	C.I. 47005	
	C.I. Acid Yellow 3	0.3000
	C.I. 16255	
	C.I. Acid red 18	0.2000
Turquoise (Blue-Green)	C.I. 42735	
	C.I. Acid Blue 104	0.4500
	C.I. 42080	
	C.I. Acid Blue 7	0.2000
	C.I. 61570	
	C.I. Acid Green 25	0.3500
Vermilion (Red-Orange)	C.I. 27290	
	C.I. Acid Red 73	0.3080
	C.I. Acid Yellow 36	0.0420
Violet	C.I. 42535	
	C.I. Basic Violet 1	0.5000
Yellow	C.I. 19555	
	C.I. Direct Yellow 28	0.4000
Yellow Green (Leaf-Green)	C.I. 19140	
	C.I. Acid Yellow 28	0.6250
	C.I. 42080	
	C.I. Acid Blue	0.3750

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TABLE I. CONT.

OPAQUE INKS		
White*	———	———

* The ink formulas listed (for transparent inks) will produce aqueous dyestuff solutions considered sufficiently brilliant and intense to be suitable standards for the applicable opaque inks. White is the only opaque color for which no surveillance by visual examination methods is possible. A method which has been found to be practicable is the ruling of several lines of 0.0076 cm (0.003 inch) and 0.050800 cm (0.0200 inch) width on a high-contrast blueprint or black board.

3.8 Workmanship. Inks shall be smooth homogeneous liquids, free from lumps, strings, and objectionable odors. The containers shall have no defects that affect serviceability or appearance.

3.9 Shelf Life. The ink shall be able to meet all the requirements of this specification from the time of delivery to a minimum of 24 months in storage under normal storage conditions.

4. Regulatory Requirements:

The offerer/contractor is encouraged to use recovered materials to the maximum extent practicable in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.1 Toxicity. The use of benzene or carbon tetrachloride solvents and any known carcinogens as ingredients is are prohibited. Trace amounts of benzene derivatives present in commercial grades of acceptable aromatics are permissible. Inks shall contain no chlorinated compounds or other toxic hydrolyzable chlorine derivatives. Use of any toxic substance must be in accordance with all applicable federal statutes.

4.2 Labeling. Stamp pad ink containers shall be labeled to comply with the "Federal Hazardous Substances Act Regulations", 16 Code of Federal Regulations Part 1500.

4.3 Material Safety Data Sheets. Safety data sheets shall be furnished in accordance with Fed-Std-313. The pertinent government mailing addresses for submission of data are listed in appendix B of Fed. Std. 313.

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5. Quality Assurance Requirements:

5.1 Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and therefore as may be otherwise provided for under the provisions of the contract.

5.2 Examination and Testing.

5.2.1 Suppliers are encouraged to use quality control (QC) techniques that exhibit control over their processes (e.g. (Statistical Process Control) SPC Techniques) as defined in American National Standards Institute (ANSI) Z1.1, Z1.2 and Z1.3 / American Society for Quality Control (ASQC) B.1, B.2 and B.3 that systematically reduce excess variations. If used, objective evidence shall be available that demonstrates overall measurement adequacy techniques and controls. These techniques shall ensure quality levels equal to, or greater than, those cited in applicable technical document or, herein.

5.2.2 End item inspection / testing may be used by the offeror or the Government representative, as a means to determine the effectiveness of the in process quality controls. In process controls shall not be substituted for end item performance testing.

5.2.3 For those characteristics for which there are no such controls, or for end item performance test, lot by lot sampling for inspection / test shall be required. The sample unit shall be one bottle of ink. Sampling shall be in accordance with the American National Standards Institute (ANSI) / American Society for Quality Control (ASQC) Z1.4, Sampling Procedures and Tables for Inspection by Attributes.

5.2.4 An inspection lot shall consist of all like items submitted for inspection at one time. The inspection level for sampling shall be S-4 with an AQL of 4.0 percent defective. The supplier must provide objective evidence (tests and inspection records) that the presented material meets the requirements of the sampling plan as indicated above.

5.3 Performance Test:

5.3.1 Determining Transparency or Opacity of Ink. Draw lines .32 cm (1/8 inch) wide with black, waterproof drawing ink on white paper. With a brush, paint strips 1.3 cm (1/2 inch) wide with each color over the black lines. If the black lines show clearly through the colors, the ink shall be classified as type I, (transparent dye ink). If the color substantially obliterates the black lines, the ink shall be classified as type II, (opaque pigment inks).

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6. Packaging:

6.1 Preservation, packing, and marking shall be specified in the contract or order.

7. Notes:

7.1 Applicable Documents. Issues of the following documents, in effect on the date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

7.1.1 Copies of ASTM test methods may be obtained at 1916 Race Street, Philadelphia, PA 19103.

7.1.2 Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, Government Printing Office, Washington. DC 20402.

7.1.3 ANSI / ASQC Z1.4 may be obtained from the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Avenue, Milwaukee, Wisconsin 53201-4606.

7.1.4 Federal Standards and Specifications may be obtained from the General Services Administration Specifications Section (3FBO-W) in Suite 8100 at 490 L'Enfant Plaza, SW Washington DC 20407.

7.2 Ordering Information. Purchaser should select the required options permitted herein, and include the following information in procurement documents.

A. Title, number, and date of this commercial item description.
B. Purchasers shall specify the type and color of ink required.
C. Purchasers shall specify size of container.
D. Purchasers shall specify arrangement for inspection and inspection facilities, if other than specified (see tests and quality assurance requirements).
E. Purchasers shall specify packaging, packing and special marking
F. Purchasers shall specify if bar code marking is not required.
G. Purchasers may specify alternative applicator style, if required.
H. MSD's required, in addition to those required by FED-STD-313.
I. Purchasers shall specify palletization, if required.

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7.3 National Stock Number Listing.

7510-00-161-4256
7510-00-233-0546
7510-00-233-0570
7510-00-237-7040
7510-00-237-7051
7510-00-238-7181
7510-00-254-2955
7510-00-254-2956
7510-00-254-2957
7510-00-254-2958
7510-00-254-2959
7510-00-254-2960
7510-01-035-8130
7510-01-035-8131
7510-01-035-8132
7510-01-035-8133
7510-01-036-3725
7510-01-036-3726

MILITARY INTERESTS:
DoD has waived coordination

PREPARING ACTIVITY
GSA - FSS

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